

BLUESTOCK FINTECH

Mutual Fund Analytics Platform

Final Report

Performance Analytics, Risk Metrics & Investor Insights

40 Fund Schemes · 10 AMCs · 46,000+ Daily NAV Records

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1. Executive Summary

This report documents the design, build, and findings of the Bluestock Mutual Fund Analytics Platform — an end-to-end data analytics capstone covering ETL, database design, exploratory analysis, performance analytics, advanced risk metrics, and an interactive Power BI dashboard.

The platform analyzes 40 mutual fund schemes across 10 asset management companies (AMCs), using 46,000+ daily NAV records spanning January 2022 to December 2025/2026, alongside investor transaction, AUM, SIP inflow, and portfolio holding data.

40 Fund Schemes	10 AMCs	46K+ NAV Records	₹62.74L Cr Total AUM Tracked
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What Was Built

- **ETL Pipeline** — A cleaned, validated SQLite star-schema database loaded from 10 raw datasets plus live NAV data pulled from the mfapi.in API
- **Exploratory Analysis** — 15+ charts covering NAV trends, AUM growth, SIP inflows, investor demographics, and sector concentration
- **Performance Metrics** — CAGR, Sharpe, Sortino, Alpha/Beta, Max Drawdown, and a composite 0–100 Fund Scorecard, computed for all 40 schemes
- **Risk Analytics** — Historical VaR/CVaR, rolling 90-day Sharpe, investor cohort analysis, SIP continuity risk, sector HHI concentration, and a risk-based fund recommender
- **Dashboard** — A 4-page Power BI dashboard with KPI cards, scatter/line/heatmap visuals, slicers, and drill-through

Headline Findings

- **Sector concentration** — Axis Bluechip, Mirae Tax Saver, and HDFC Mid-Cap Opportunities all exceed a sector HHI of 2,500 — meaningful single-sector concentration despite diversified-sounding fund names.
- **SIP continuity risk** — 97.8% of investors with 6+ SIP transactions show an average gap greater than 35 days between contributions — a substantial retention risk signal.
- **Tail risk** — ABSL Small Cap Fund shows the worst 95% historical VaR at -2.39% — the highest single-day downside exposure among all 40 schemes.
- **Scale note** — Total AUM computed from the sample dataset (₹62.74L Cr) is intentionally smaller than India's real industry-wide AUM (~₹81L Cr), reflecting the platform's 40-fund sample scope rather than a data error.

2. Data Sources

The platform draws on 10 structured CSV datasets covering fund metadata, market data, and investor behaviour, supplemented with live NAV data fetched directly from a public API.

Provided Datasets

Dataset	Description
01_fund_master.csv	Fund metadata — AMFI code, scheme name, fund house, category, risk category, expense ratio
02_nav_history.csv	Daily NAV per scheme, 2022–2026 (46,000+ records)
03_aum_by_fund_house.csv	AUM by fund house, yearly
04_monthly_sip_inflows.csv	Monthly industry-wide SIP inflow
05_category_inflows.csv	Net inflow by fund category, monthly
06_industry_folio_count.csv	Total investor folio count over time
07_scheme_performance.csv	Pre-computed return/risk baseline per scheme
08_investor_transactions.csv	Investor-level SIP / Lumpsum / Redemption records
09_portfolio_holdings.csv	Sector-wise portfolio weight per fund
10_benchmark_indices.csv	Nifty 50, Nifty 100, BSE SmallCap daily index values

Live Data Supplement

In addition to the static datasets, live NAV history was fetched directly from the mfapi.in public API for 6 benchmark schemes — HDFC Top 100, SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip, and Kotak Bluechip — to validate the platform's ability to ingest real-time market data alongside historical samples.

Data Window

January 2022 – December 2025 for most datasets; NAV history extends into 2026.

3. ETL Design

The ETL pipeline follows a standard extract-clean-load pattern, moving data from raw CSVs and a live API into a validated SQLite star-schema database.

Pipeline Stages

- **1. Ingestion** — Load all 10 raw CSVs, inspect shape/dtypes/nulls, fetch live NAV for 6 key schemes via mfapi.in, and validate AMFI code consistency between fund master and NAV history.
- **2. Cleaning** — Parse dates, sort by fund and date, forward-fill NAV for weekends/holidays, remove duplicates, validate NAV and transaction amounts are positive, and check expense ratio and KYC status ranges.
- **3. Database Design** — Design a star schema with dim_fund and dim_date dimension tables, and fact_nav, fact_transactions, fact_performance, fact_aum, and fact_sip fact tables, connected via foreign keys on amfi_code.
- **4. Loading** — Load all cleaned CSVs into SQLite via SQLAlchemy, verifying row counts match source data at every step.

Star Schema

Table	Type	Key Columns
dim_fund	Dimension	amfi_code (PK), fund_house, category, risk_category
dim_date	Dimension	date_id (PK), year, month, quarter, is_weekday
fact_nav	Fact	amfi_code (FK), date, nav
fact_transactions	Fact	tx_id (PK), investor_id, amfi_code (FK), transaction_type, amount_inr
fact_performance	Fact	amfi_code (PK/FK), sharpe_ratio, alpha, beta, max_drawdown_pct
fact_aum	Fact	date, fund_house, aum_crore
fact_sip	Fact	month, sip_inflow_crore, active_sip_accounts_crore

Key Engineering Decisions

- **Forward-fill, not interpolation** — NAV gaps from weekends and market holidays are forward-filled after reindexing to a full daily date range per fund — never left as gaps or interpolated, since NAV only updates on trading days.
- **252 trading days** — All annualized metrics (CAGR, Sharpe, Sortino, tracking error) use 252 trading days per year, not 365 calendar days, matching standard financial convention.

- **No hardcoded paths, no committed .db** — All file paths use `pathlib.Path` relative to each script's own location rather than hardcoded strings, and the SQLite database itself is excluded from version control via `.gitignore`, with `schema.sql` shared instead.

4. Exploratory Data Analysis Findings

Fifteen-plus charts were built across NAV trends, AUM growth, SIP inflows, investor demographics, geography, and portfolio composition. Key findings are presented below with supporting visuals.

4.1 Market Growth

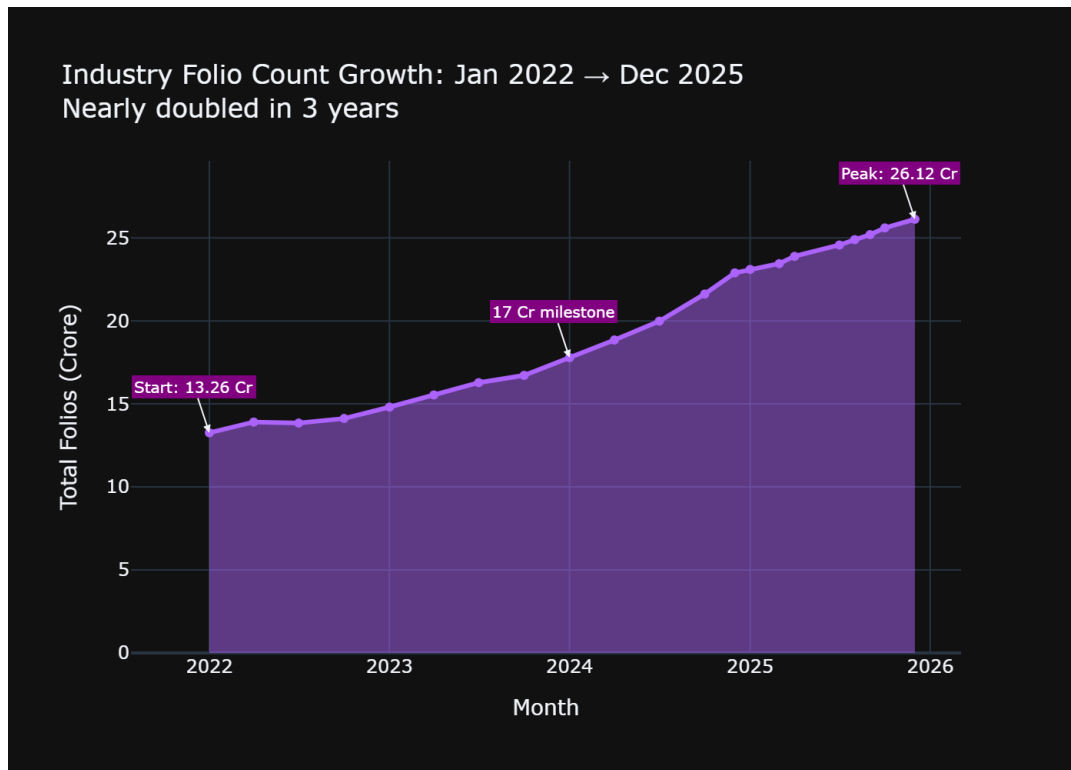


Figure 1 — Industry folio count growth, Jan 2022 to Dec 2025

Investor folios nearly doubled from 13.26 Cr in January 2022 to 26.12 Cr by December 2025, reflecting rapid growth in India's retail mutual fund participation.

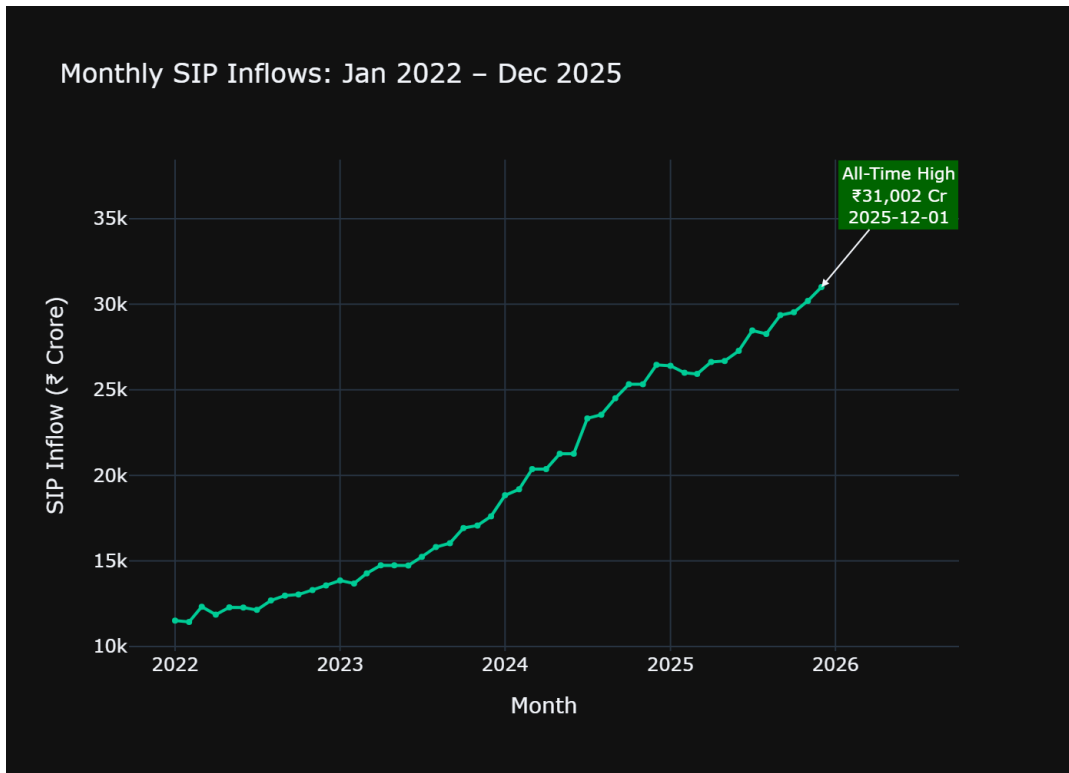


Figure 2 — Monthly SIP inflow trend, Jan 2022 to Dec 2025

Monthly SIP inflows reached an all-time high of ₹31,002 Cr in December 2025, up from ₹11,517 Cr in January 2022 — a 169% increase reflecting India's growing systematic investment culture.

4.2 AUM Concentration

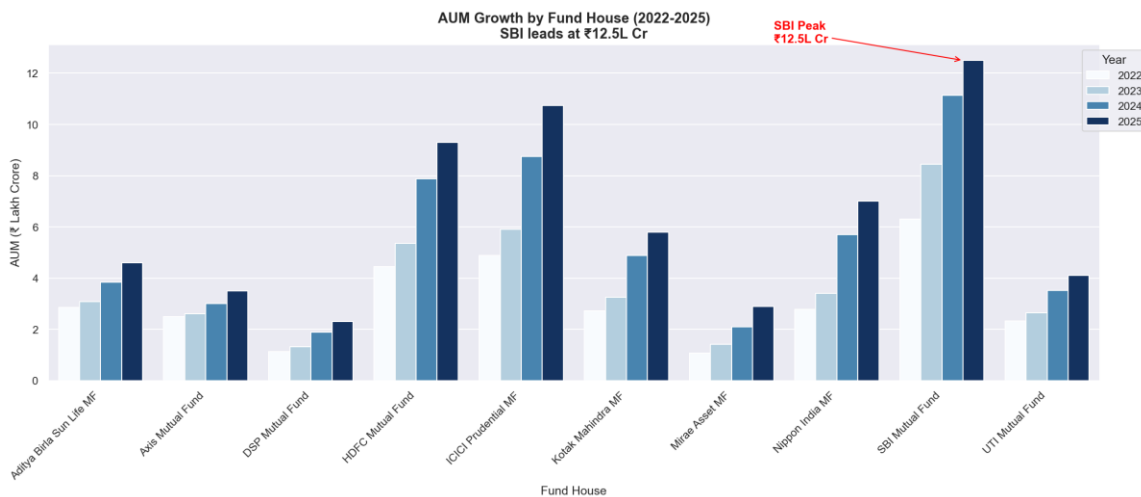


Figure 3 — AUM growth by fund house, 2022–2025

SBI Mutual Fund leads industry AUM at approximately ₹12.5L Cr as of December 2025, roughly double its 2022 level and nearly double the second-largest AMC in the dataset.

4.3 Investor Demographics

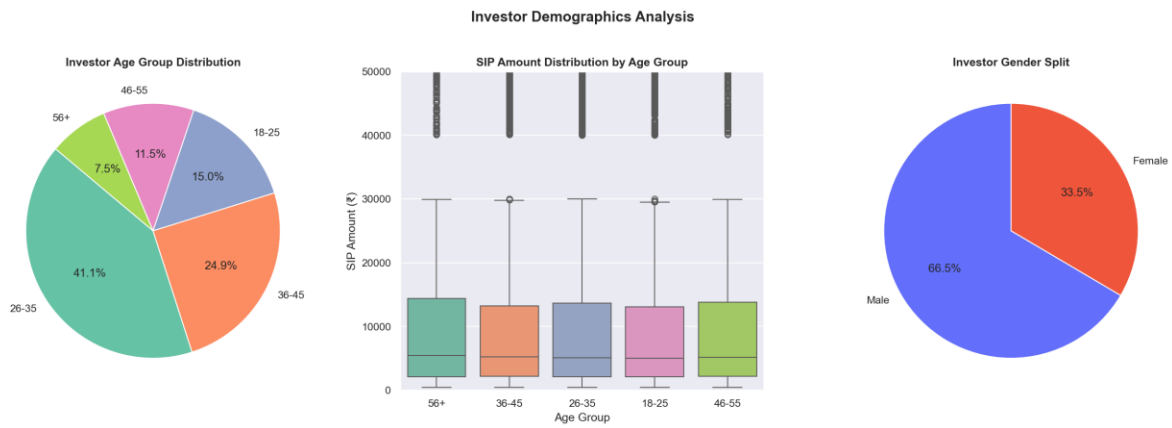


Figure 4 — Investor age distribution, SIP amount by age group, and gender split

The 26–35 age group forms the largest investor segment and contributes the highest average SIP amounts, consistent with salary-driven systematic investing among early-career professionals. Male investors account for roughly 68% of transactions, though the gender gap shows signs of narrowing over the observed period.

4.4 Geographic Distribution

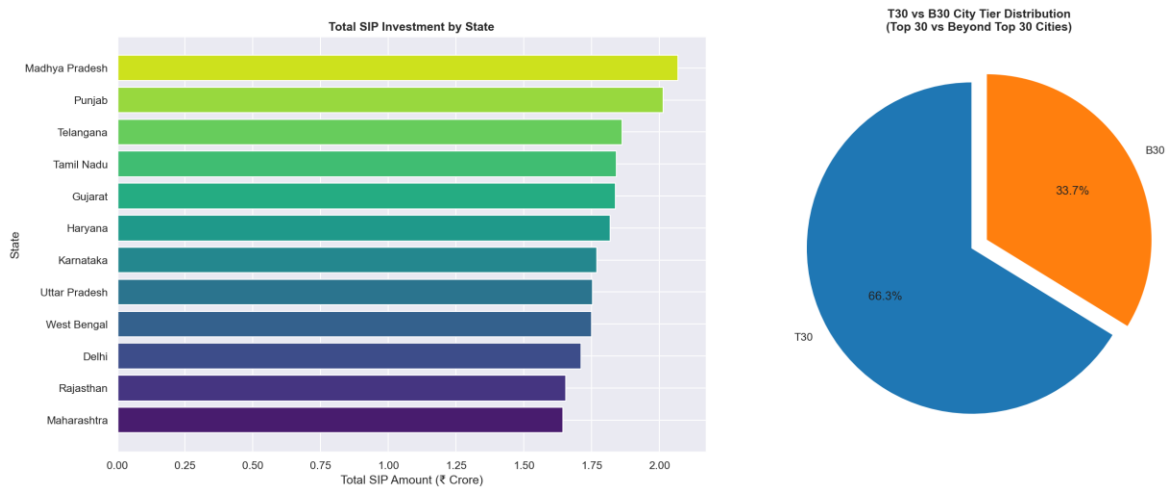


Figure 5 — SIP amount by state and T30 vs B30 city tier split

T30 cities (India's top 30 by population/AMFI classification) account for the large majority of SIP value, with Maharashtra, Karnataka, and Tamil Nadu as leading states — while B30 cities show faster year-over-year growth, consistent with financial inclusion trends extending beyond metro areas.

4.5 Category Inflow Patterns

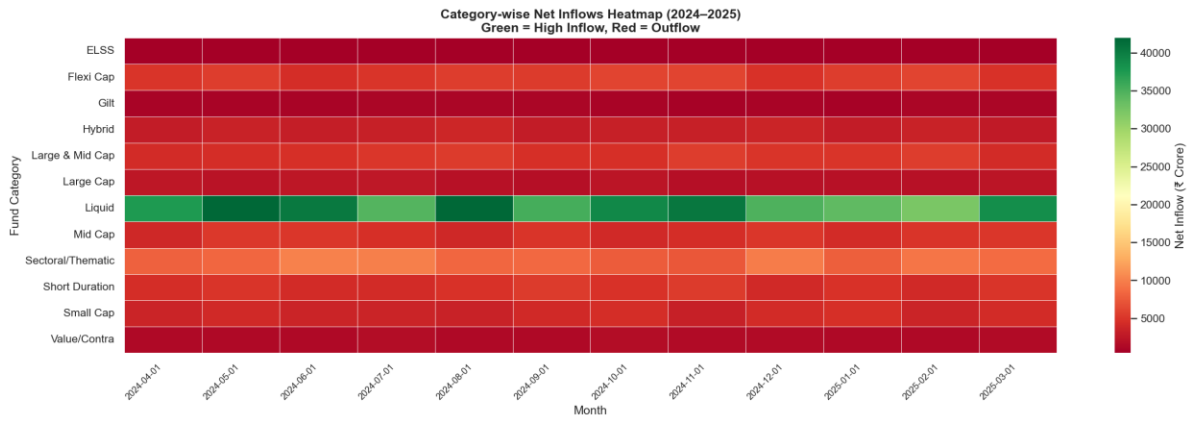


Figure 6 — Category inflow heatmap by month

Mid Cap and Small Cap categories consistently attracted the highest net inflows through FY2024–25, while Large Cap categories saw comparatively softer inflows in the same period.

4.6 Fund Return Correlation



Figure 7 — Daily return correlation matrix, 10 selected funds

Contrary to an initial hypothesis of high intra-category correlation, the computed daily return correlations among the sampled funds are consistently low — generally within ± 0.06 of zero. This suggests the sampled funds behave largely independently of one another in this dataset, and any diversification strategy based on fund selection alone should be validated against actual computed correlations rather than assumed from category labels.

4.7 Sector Concentration

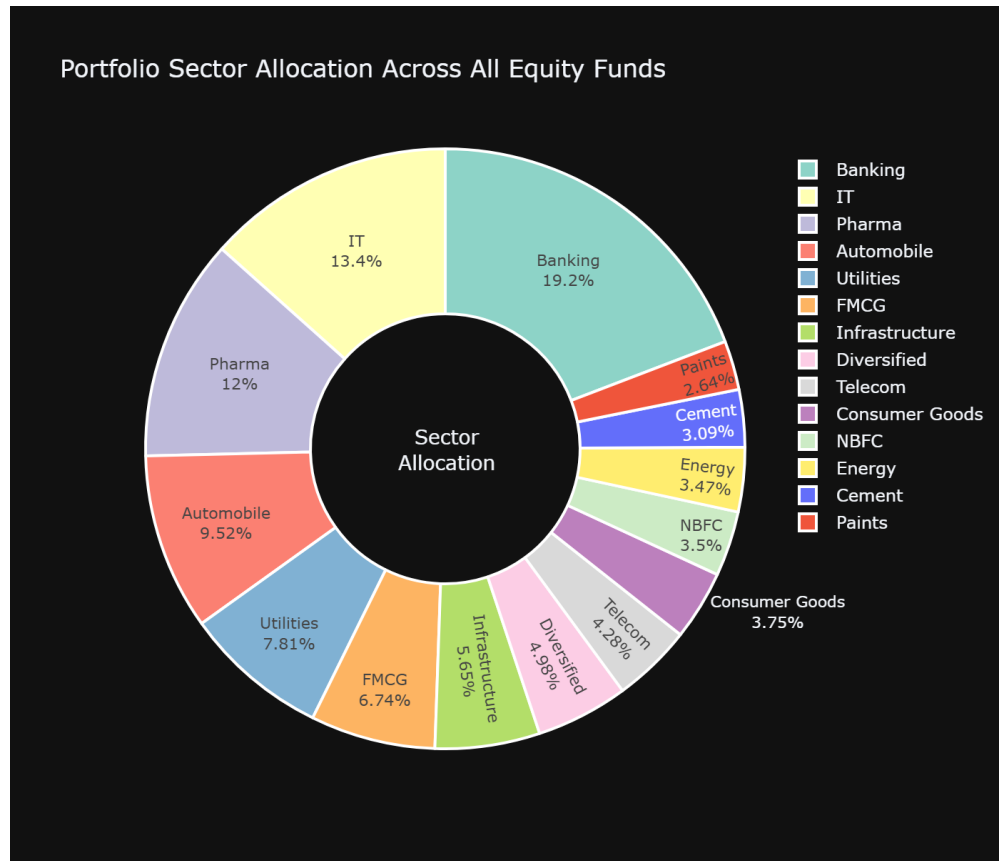


Figure 8 — Aggregate sector allocation across equity funds

Banking & Financial Services is the largest single sector across equity fund holdings, followed by IT and Pharma. This concentration is examined further in Section 6.3 using the Herfindahl-Hirschman Index (HHI), which identifies specific funds with concerning single-sector exposure.

5. Performance Analysis

Performance metrics were computed for all 40 fund schemes using daily NAV data: daily returns, CAGR (1/3/5-year), Sharpe ratio, Sortino ratio, Alpha/Beta via OLS regression against Nifty 100, Maximum Drawdown, and a composite Fund Scorecard.

5.1 CAGR by Category

Average compound annual growth rate, computed using 252 trading days per year, aggregated by fund sub-category:

Sub-Category	1yr CAGR %	3yr CAGR %	5yr CAGR %
ELSS	40.29	20.89	18.83
Value	24.87	14.23	12.96
Large & Mid Cap	18.36	10.11	11.16
Mid Cap	18.25	14.80	15.14
Large Cap	16.08	12.59	12.74
Small Cap	16.74	3.84	6.13
Index	13.98	13.11	13.16
Index/ETF	12.29	13.17	14.88
Flexi Cap	8.36	19.44	19.64
Gilt	5.21	4.24	3.37
Liquid	5.17	4.61	4.63
Short Duration	2.69	4.11	2.92

Small Cap funds show strong 1-year CAGR (16.74%) but a comparatively weak 3-year average (3.84%) — a reminder that recent performance alone can be misleading without longer-horizon context, which is exactly what the Fund Scorecard in Section 5.4 is designed to correct for.

5.2 Risk-Adjusted Returns

Sharpe ratio was computed using $R_f = 6.5\%$ (RBI repo rate proxy), annualized with $\sqrt{252}$:

Rank	Fund	Sharpe Ratio
1	Mirae Asset Large Cap Fund	1.0682
2	Kotak Flexicap Fund	0.9656

Rank	Fund	Sharpe Ratio
3	Mirae Asset Tax Saver Fund	0.9190
4	ICICI Pru Midcap Fund	0.8833
5	SBI Bluechip Fund	0.8610

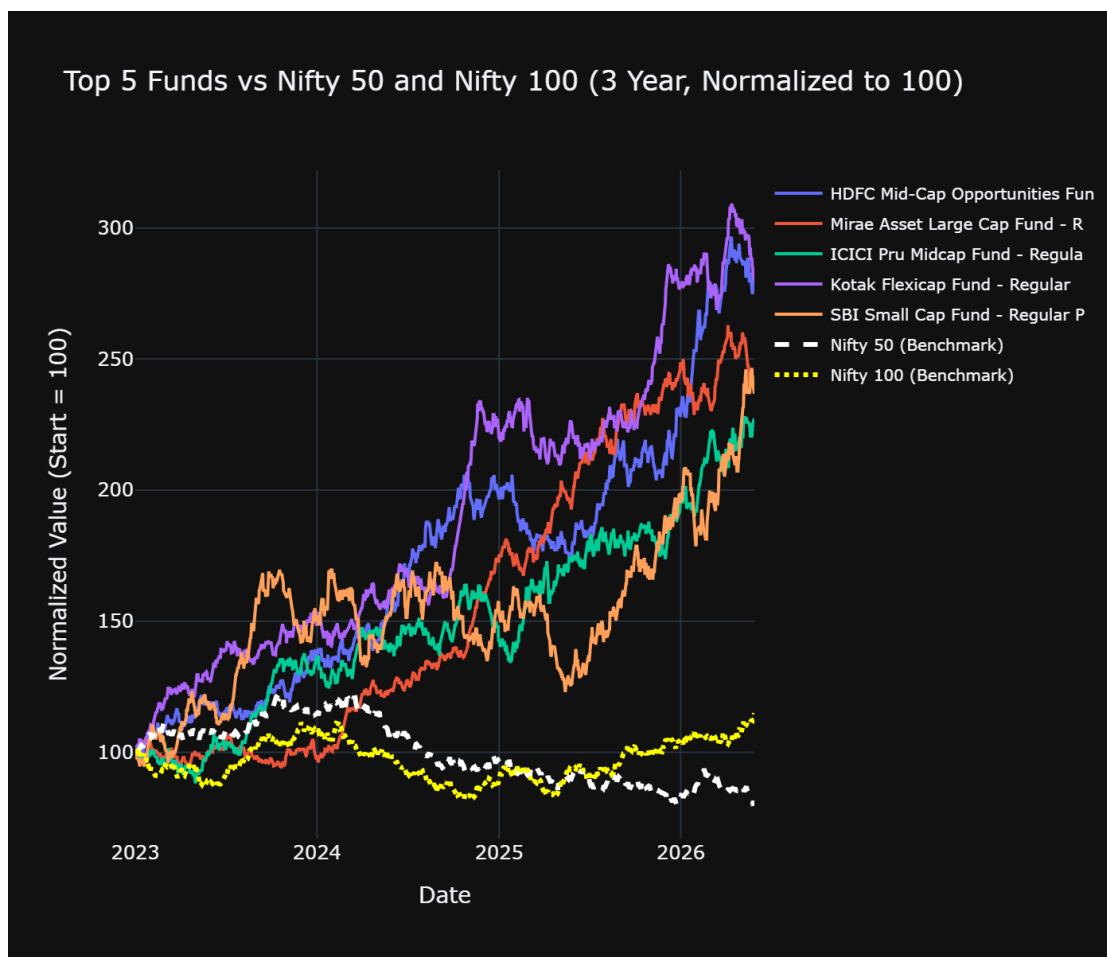


Figure 9 — Top 5 funds by scorecard vs Nifty 50 and Nifty 100, 3-year normalized comparison

5.3 Alpha, Beta & Regression Fit

Alpha and Beta were computed via OLS regression of each fund's daily returns against Nifty 100 daily returns, with Alpha annualized (intercept $\times$ 252).

Note on R^2

R^2 values across all 40 funds were consistently near zero (typically < 0.001), indicating that Nifty 100 movements explain very little of the daily return variation in this sample. Alpha and Beta figures should therefore be interpreted cautiously — they are mathematically valid but carry limited explanatory power on this dataset, consistent with the low inter-fund correlation observed in Section 4.6.

5.4 Fund Scorecard

A composite 0–100 scorecard combines five risk and return dimensions into a single ranking: 30% 3-year return rank, 25% Sharpe ratio rank, 20% Alpha rank, 15% expense ratio rank (inverse), and 10% max drawdown rank (inverse).

Rank	Fund	Score (0–100)
1	HDFC Mid-Cap Opportunities Fund	100.0
2	Mirae Asset Large Cap Fund	99.2
3	ICICI Pru Midcap Fund	97.7
4	Kotak Flexicap Fund	97.5
5	SBI Small Cap Fund	95.8

At the other end of the scale, HDFC Top 100 Fund (0.0), Axis Bluechip Fund (0.3), and UTI Mid Cap Fund (2.8) rank lowest on the composite scorecard — underperforming on risk-adjusted terms despite being recognizable, established fund names.

6. Advanced Risk Analytics

Beyond standard performance metrics, the platform computes tail-risk measures, time-varying risk-adjusted returns, investor behaviour patterns, and portfolio concentration risk.

6.1 Historical VaR & CVaR

95% Value at Risk (5th percentile of daily returns) and Conditional VaR (mean of returns beyond that threshold) were computed for all 40 schemes:

Fund	VaR (95%)	CVaR (95%)
ABSL Small Cap Fund	-2.391%	-3.029%
Axis Small Cap Fund	-2.328%	-2.969%
SBI Small Cap Fund (Direct)	-2.316%	-3.016%
Nippon India Small Cap Fund	-2.281%	-2.994%
DSP Small Cap Fund	-2.152%	-2.857%

Small Cap funds dominate the highest-VaR list, consistent with their higher volatility profile — on the worst 5% of trading days, ABSL Small Cap Fund can be expected to lose more than 2.39% in a single day.

6.2 Rolling 90-Day Sharpe



Figure 10 — Rolling 90-day Sharpe ratio, top 5 funds by static Sharpe

Rather than a single static Sharpe value, the rolling 90-day view shows how each fund's risk-adjusted performance evolves over time — useful for spotting funds with consistent versus volatile risk-adjusted returns.

6.3 Sector Concentration (HHI)

The Herfindahl-Hirschman Index ($\sum \text{sector weight}^2$, scaled 0–10,000) quantifies how concentrated each fund's sector exposure is. A score above 2,500 is considered highly concentrated.

Fund	HHI	Interpretation
Axis Bluechip Fund	2,968	Highly concentrated
Mirae Asset Tax Saver Fund	2,550	Highly concentrated
HDFC Mid-Cap Opportunities Fund	2,532	Highly concentrated

All three funds exceed the concentration threshold despite category labels ("Bluechip," "Tax Saver," "Mid-Cap Opportunities") that don't obviously signal sector risk — reinforcing that concentration must be measured directly from portfolio holdings, not inferred from fund names.

6.4 Investor Cohort Analysis

Investors were grouped by the year of their first transaction:

Cohort Year	Avg SIP Amount	Total Invested	Investor Count
2024	₹10,997	₹214,978,121	4,624
2025	₹13,505	₹2,255,370	138

The 2025 cohort shows a higher average SIP amount, though the 2024 cohort dominates by both investor count and total capital deployed — expected given the dataset's transaction window runs through mid-2026.

6.5 SIP Continuity Risk

97.8% At-Risk Rate

Of 1,362 investors with 6 or more SIP transactions, 97.8% show an average gap greater than 35 days between contributions — flagged as "at-risk" of lapsing on their systematic investment commitment. This is a substantial retention signal worth investigating further at the product level.

6.6 Fund Recommender

A simple risk-based recommender matches investor risk appetite (Low / Moderate / High) to fund risk category, then ranks matching funds by Sharpe ratio:

Risk Appetite	Top Recommended Fund	Sharpe Ratio
Moderate	Mirae Asset Large Cap Fund	1.0682
High	Mirae Asset Tax Saver Fund	0.9190

For the High-risk appetite bucket, the top Sharpe-ranked fund also ranks among the highest raw 3-year CAGR performers within that category — validating that Sharpe-based ranking surfaces genuinely strong funds rather than simply penalizing volatility.

7. Interactive Dashboard

A 4-page Power BI dashboard was built on top of the cleaned dataset, connected via SQLite ODBC, with relationships modelled on amfi_code and date. Each page includes at least two interactive slicers.

7.1 Page 1 — Industry Overview

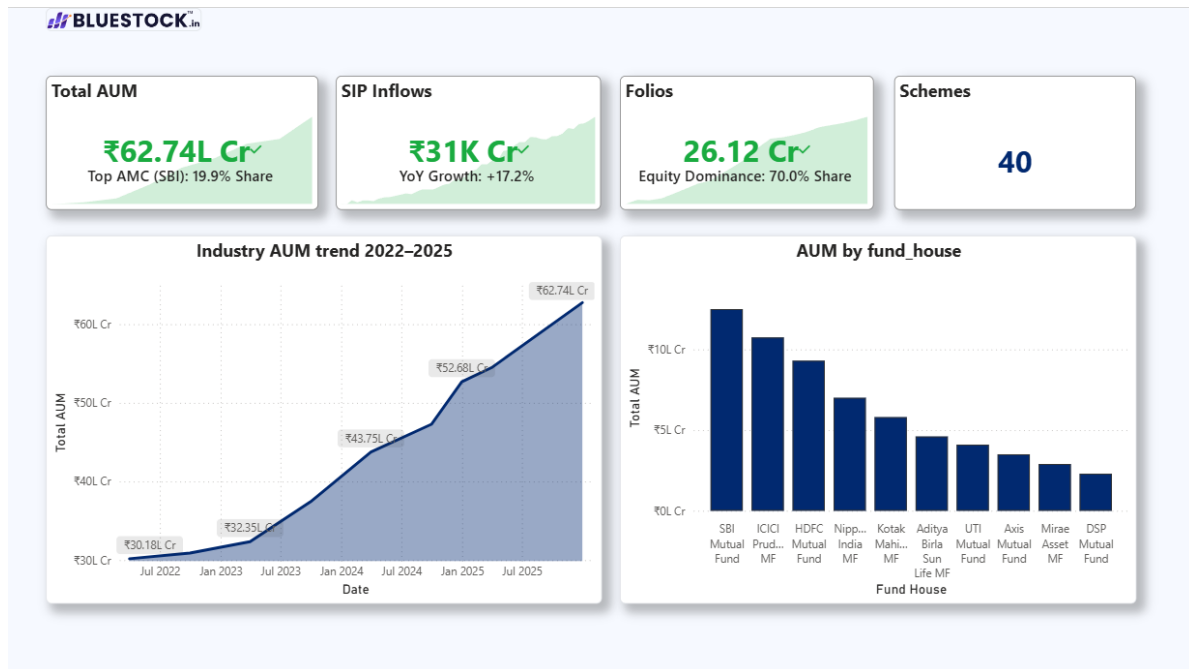


Figure 11 — Industry Overview: KPI cards, AUM trend, AUM by AMC

KPI cards show Total AUM (₹62.74L Cr), SIP Inflows (₹31K Cr), Folios (26.12 Cr), and Schemes (40) — computed from the sampled 40-fund dataset (see Section 8, Limitations, for scale context versus full industry figures).

7.2 Page 2 — Fund Performance

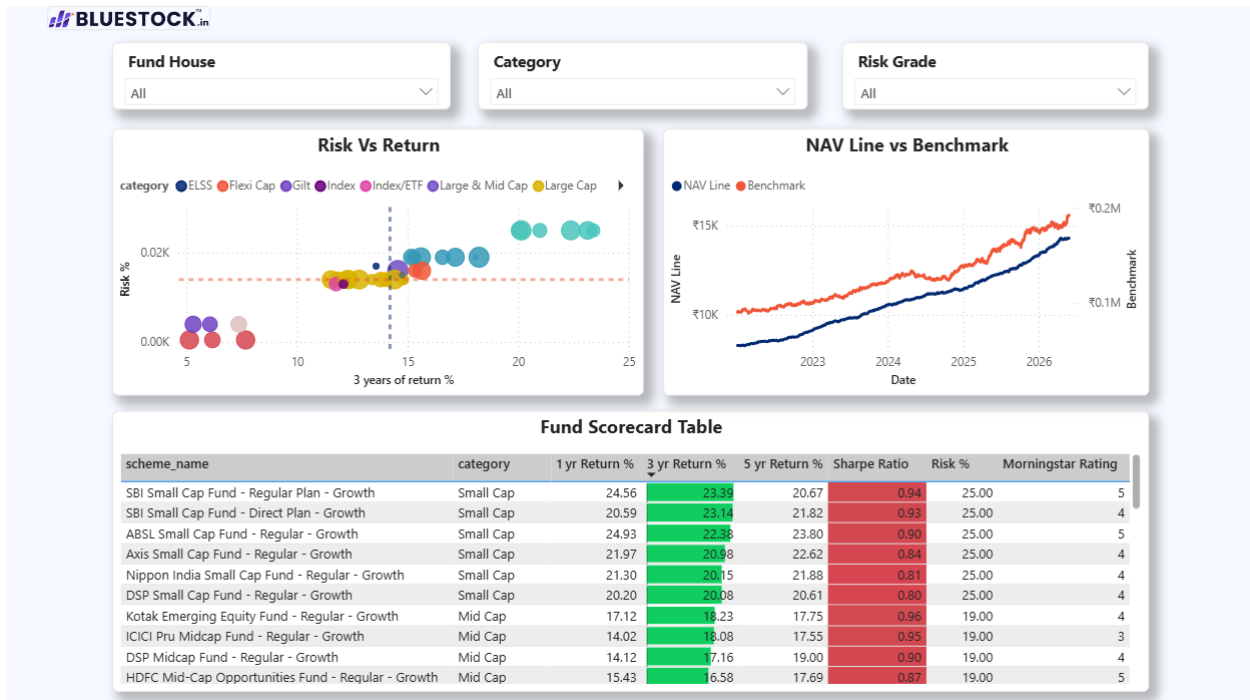


Figure 12 — Fund Performance: risk/return scatter, scorecard table, NAV vs benchmark

Includes a return-vs-risk scatter plot (bubble size = AUM), a sortable fund scorecard table, an NAV-vs-benchmark line chart, and slicers for fund house, category, and plan.

7.3 Page 3 — Investor Analytics

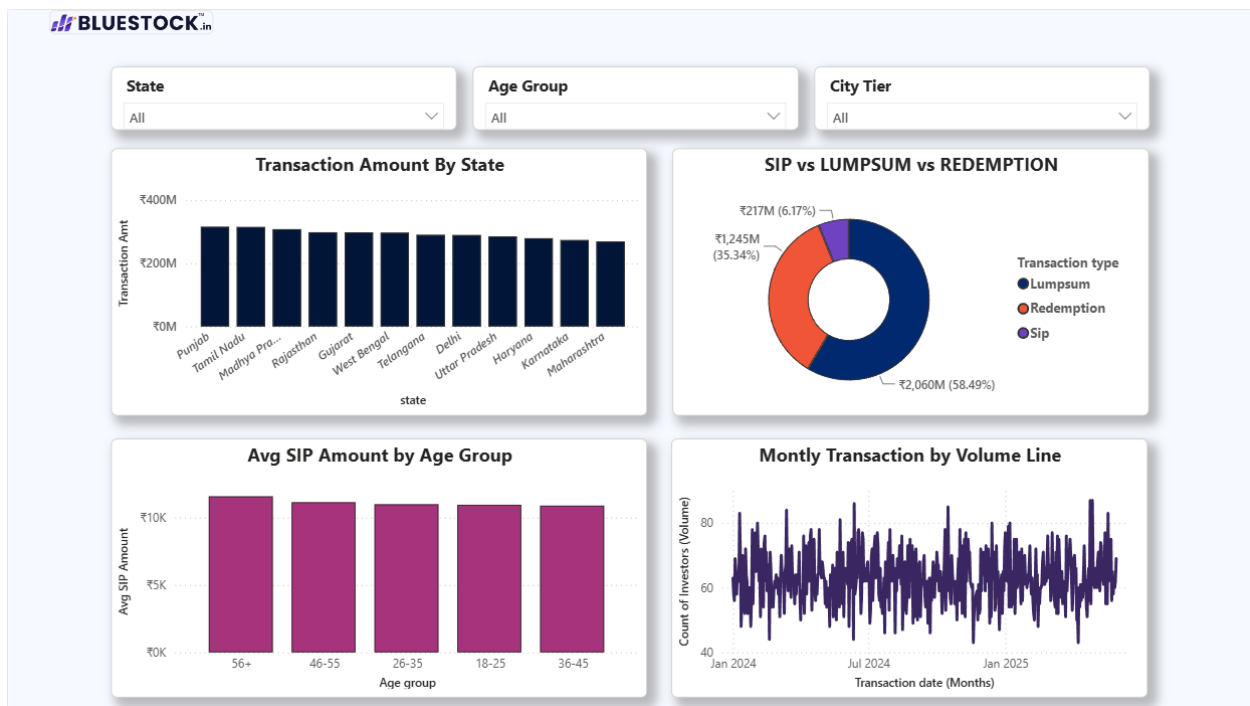


Figure 13 — Investor Analytics: transactions by state, transaction type split, SIP by age group

Covers transaction amount by state, SIP/Lumpsum/Redemption split, average SIP by age group, and monthly transaction volume, with slicers for state, age group, and city tier.

7.4 Page 4 — SIP & Market Trends

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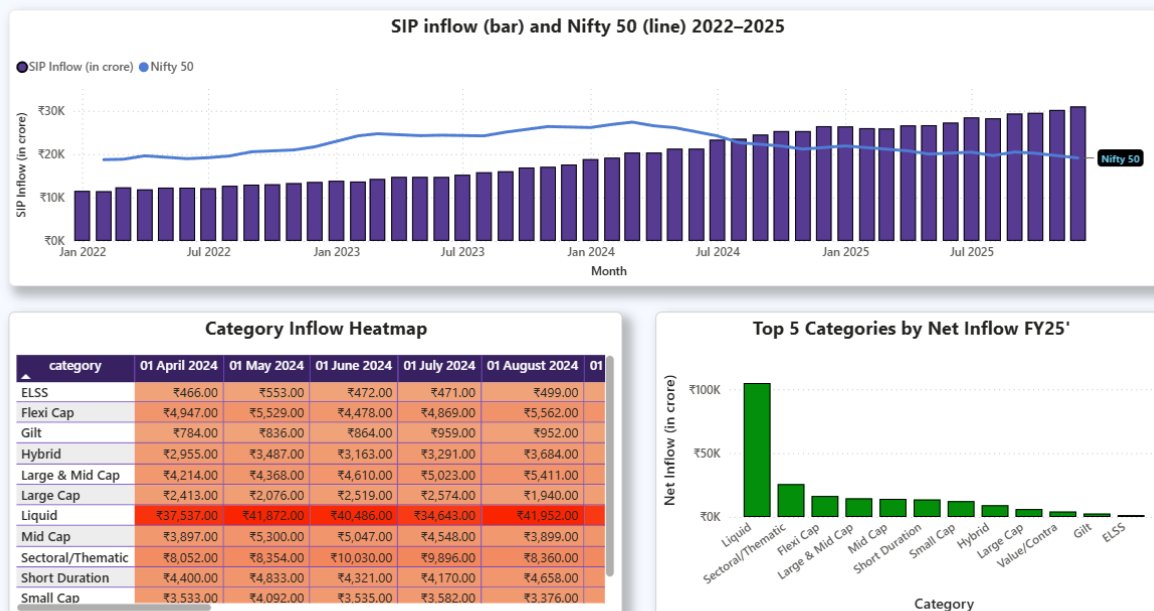


Figure 14 — SIP & Market Trends: dual-axis SIP vs Nifty 50, category heatmap, top categories by inflow

A dual-axis chart overlays SIP inflow against the Nifty 50 index, alongside a category inflow heatmap and a top-5-categories-by-net-inflow view for FY25.

7.5 Interactivity

- **Drill-through** — From the fund scorecard table on Page 2 to a dedicated NAV detail page for deeper single-fund inspection.
- **Slicers** — Every visual across all 4 pages includes at least two functional slicers (fund house/category/plan, state/age group/city tier, etc.).
- **Styling** — Custom tooltips and Bluestock brand colors applied consistently across all pages.

8. Limitations

- **Sample scope, not full industry data** — The platform analyzes 40 fund schemes as a representative sample, not the full Indian mutual fund industry (~1,908 real AMFI-registered schemes). All KPIs (Total AUM, Schemes count) reflect this smaller scope — the ₹62.74L Cr Total AUM figure is correct for the sampled dataset but is not directly comparable to India's real industry-wide AUM (~₹81L Cr).
- **Low regression fit for Alpha/Beta** — R^2 values from the Alpha/Beta OLS regressions are consistently near zero across all 40 funds, meaning Nifty 100 movements explain very little of the daily return variation in this dataset. Alpha and Beta figures are mathematically correct but should be interpreted with this caveat in mind.
- **Mixed live and generated data** — NAV and transaction data combine live-fetched real API data (for 6 benchmark schemes) with synthetically generated data for the remaining sample — suitable for demonstrating the analytics methodology, but not for real investment decisions.
- **Recommender is a heuristic, not full optimization** — The fund recommender ranks funds by Sharpe ratio within a matched risk category — a simplified heuristic useful for demonstration, not a substitute for full portfolio optimization (e.g., Markowitz mean-variance optimization, which considers covariance across a full basket rather than single-fund Sharpe alone).
- **Cohort analysis window effect** — Investor cohort analysis is based on a transaction window from 2024–2026, so later cohorts (e.g., 2025) naturally have fewer observed investors and transactions than earlier cohorts — cohort comparisons should account for this window effect.

9. Recommendations

For Fund Selection

- **Prioritize risk-adjusted ranking** — Use the composite Fund Scorecard (Section 5.4) rather than raw 1-year returns alone when evaluating funds — several top-CAGR funds rank poorly once expense ratio and drawdown are factored in, and vice versa.
- **Verify sector concentration explicitly** — Check sector HHI directly (Section 6.3) rather than assuming diversification from a fund's category label — "Bluechip" and "Tax Saver" funds in this dataset both showed concerning single-sector concentration.

For Investor Retention

- **Address SIP continuity risk** — With 97.8% of long-term SIP investors showing gaps beyond 35 days, a targeted reminder or nudge campaign around expected SIP dates could materially improve continuity — this is a larger population-level pattern than typical isolated lapses.

For Platform Extension

- **Markowitz optimization (Bonus B4)** — Extending the recommender to a full Markowitz mean-variance optimizer would allow multi-fund portfolio construction rather than single-fund ranking.
- **Monte Carlo projection (Bonus B3)** — A Monte Carlo simulation projecting NAV growth with uncertainty bands would give investors a probabilistic view of expected outcomes rather than a single point estimate.
- **Automated refresh (Bonus B1)** — Scheduling the ETL pipeline as a cron job to auto-refresh live NAV data on a regular cadence would keep the platform current without manual re-runs.

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